

41. $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3} \Leftrightarrow \left(\frac{a}{b}\right)^{x-1} = \left(\frac{a}{b}\right)^{-(x-3)} = \left(\frac{a}{b}\right)^{(3-x)}$
 $\Leftrightarrow x-1 = 3-x \Leftrightarrow 2x = 4 \Leftrightarrow x = 2.$
42. $2^{2n-1} = \frac{1}{8^{n-3}} \Leftrightarrow 2^{2n-1} = \frac{1}{(2^3)^{n-3}}$
 $= \frac{1}{2^{3(n-3)}} = \frac{1}{2^{3n-9}} = 2^{(9-3n)}$
 $\Leftrightarrow 2n-1 = 9-3n \Leftrightarrow 5n = 10$
 $\Leftrightarrow n = 2.$
43. $5^a = 3125 = 5^5 \Rightarrow a = 5. \Rightarrow 5^{(a-3)} = 5^{(5-3)} = 5^2 = 25.$
44. $5\sqrt{5} \times 5^3 \div 5^{-\frac{3}{2}} = 5^{a+2} \Leftrightarrow \frac{5 \times 5^2 \times 5^3}{5^{-\frac{3}{2}}} = 5^{a+2}$
 $\Leftrightarrow 5^{\left(1+\frac{1}{2}+3+\frac{3}{2}\right)} = 5^{a+2} \Leftrightarrow 5^6 = 5^{a+2}$
 $\Leftrightarrow a+2 = 6 \Leftrightarrow a = 4.$
45. $\sqrt{2^n} = 64 \Leftrightarrow (2^n)^{\frac{1}{2}} = 2^6 \Leftrightarrow 2^{\frac{n}{2}} = 2^6 \Leftrightarrow \frac{n}{2} = 6$
 $\Leftrightarrow n = 12.$
46. $(\sqrt{3})^5 \times 9^2 = 3^n \times 3\sqrt{3} \Leftrightarrow \left(3^{\frac{1}{2}}\right)^5 \times (3^2)^2 = 3^n \times 3 \times 3^{\frac{1}{2}}$
 $\Leftrightarrow 3^{\left(\frac{1}{2} \times 5\right)} \times 3^{(2 \times 2)} = 3^{\left(n+1+\frac{1}{2}\right)}$
 $\Leftrightarrow 3^{\left(\frac{5}{2}+4\right)} = 3^{\left(n+\frac{3}{2}\right)} \Leftrightarrow n+\frac{3}{2} = \frac{13}{2}$
 $\Leftrightarrow n = \left(\frac{13}{2} - \frac{3}{2}\right) = \frac{10}{2} = 5.$
47. $\frac{9^n \times 3^5 \times (27)^3}{3 \times (81)^4} = 27 \Leftrightarrow \frac{(3^2)^n \times 3^5 \times (3^3)^3}{3 \times (3^4)^4} = 3^3$
 $\Leftrightarrow \frac{3^{2n} \times 3^5 \times 3^{(3 \times 3)}}{3 \times 3^{(4 \times 4)}} = 3^3$
 $\Leftrightarrow \frac{3^{2n+5+9}}{3 \times 3^{16}} = 3^3 \Leftrightarrow \frac{3^{2n+14}}{3^{17}} = 3^3$
 $\Leftrightarrow 3^{(2n+14-17)} = 3^3$
 $\Leftrightarrow 3^{2n-3} = 3^3 \Leftrightarrow 2n-3 = 3 \Leftrightarrow 2n = 6 \Leftrightarrow n = 3.$
48. $\left(\frac{9}{4}\right)^x \cdot \left(\frac{8}{27}\right)^{x-1} = \frac{2}{3} \Leftrightarrow \frac{9^x}{4^x} \times \frac{8^{(x-1)}}{27^{(x-1)}} = \frac{2}{3}$
 $\Leftrightarrow \frac{(3^2)^x}{(2^2)^x} \times \frac{(2^3)^{(x-1)}}{(3^3)^{(x-1)}} = \frac{2}{3} \Leftrightarrow \frac{3^{2x} \times 2^{3(x-1)}}{2^{2x} \times 3^{3(x-1)}} = \frac{2}{3}$
 $\Leftrightarrow \frac{2^{(3x-3-2x)}}{3^{(3x-3-2x)}} = \frac{2}{3} \Leftrightarrow \frac{2^{(x-3)}}{3^{(x-3)}} = \frac{2}{3} \Leftrightarrow \left(\frac{2}{3}\right)^{(x-3)} = \left(\frac{2}{3}\right)^1$
 $\Leftrightarrow x-3 = 1 \Leftrightarrow x = 4.$

49. $2^x = \sqrt[3]{32} \Leftrightarrow 2^x = (32)^{\frac{1}{3}} = (2^5)^{\frac{1}{3}} = 2^{\frac{5}{3}} \Leftrightarrow x = \frac{5}{3}.$
50. $2^x \times 8^{\frac{1}{5}} = 2^{\frac{1}{5}} \Leftrightarrow 2^x \times (2^3)^{\frac{1}{5}} = 2^{\frac{1}{5}} \Leftrightarrow 2^x \times 2^{\frac{3}{5}} = 2^{\frac{1}{5}}$
 $\Leftrightarrow 2^{\left(x+\frac{3}{5}\right)} = 2^{\frac{1}{5}}$
 $\Leftrightarrow x+\frac{3}{5} = \frac{1}{5} \Leftrightarrow x = \left(\frac{1}{5} - \frac{3}{5}\right) = -\frac{2}{5}.$
51. $5^{(x+3)} = 25^{(3x-4)} \Leftrightarrow 5^{(x+3)} = (5^2)^{(3x-4)} \Leftrightarrow 5^{(x+3)} = 5^{2(3x-4)}$
 $= 5^{2(3x-4)} \Leftrightarrow 5^{(x+3)} = 5^{(6x-8)}$
 $\Leftrightarrow x+3 = 6x-8 \Leftrightarrow 5x = 11$
 $\Leftrightarrow x = \frac{11}{5}.$
52. $\frac{2^{n+4} - 2(2^n)}{2(2^{n+3})} = \frac{2^{n+4} - 2^{n+1}}{2^{n+4}} = \frac{2^{n+4}}{2^{n+4}} - \frac{2^{n+1}}{2^{n+4}}$
 $= 1 - 2^{n+1-(n+4)} = 1 - 2^{-3} = 1 - \frac{1}{8} = \frac{7}{8}.$
53. $\left[\sqrt[3]{6\sqrt{a^9}}\right]^4 \left[\sqrt[6]{3\sqrt{a^9}}\right]^4 = \left[\left\{\left(a^9\right)^{\frac{1}{6}}\right\}^{\frac{1}{3}}\right]^4 \cdot \left[\left\{\left(a^9\right)^{\frac{1}{3}}\right\}^{\frac{1}{6}}\right]^4$
 $= a^{\left(9 \times \frac{1}{6} \times \frac{1}{3} \times 4\right)} \cdot a^{\left(9 \times \frac{1}{3} \times \frac{1}{6} \times 4\right)}$
 $= a^2 \cdot a^2 = a^4.$
54. $(256)^{0.16} \times (256)^{0.09} = (256)^{(0.16+0.09)} = (256)^{0.25}$
 $= (256)^{\left(\frac{25}{100}\right)}$
 $= (256)^{\frac{1}{4}} = (4^4)^{\frac{1}{4}} = 4^{\left(4 \times \frac{1}{4}\right)} = 4^1 = 4.$
55. $(0.04)^{-1.5} = \left(\frac{4}{100}\right)^{-1.5} = \left(\frac{1}{25}\right)^{-\frac{3}{2}} = (25)^{\frac{3}{2}} = (5^2)^{\frac{3}{2}}$
 $= 5^{\left(2 \times \frac{3}{2}\right)} = 5^3 = 125.$
56. Let $(17)^{3.5} \times (17)^x = 17^8$. Then, $(17)^{(3.5+x)} = (17)^8$.
 $\therefore 3.5 + x = 8 \Leftrightarrow x = (8 - 3.5)$
 $\Leftrightarrow x = 4.5.$
57. Let $6^{1.2} \times 36^x \times 30^{2.4} \times 25^{1.3} = 30^5$.
Then, $6^{1.2} \times (6^2)^x \times (6 \times 5)^{2.4} \times (5^2)^{1.3} = 30^5$
 $\Leftrightarrow 6^{1.2} \times 6^{2x} \times 6^{2.4} \times 5^{2.4} \times 5^{2.6} = (6 \times 5)^5$
 $\Leftrightarrow 6^{(1.2+2x+2.4)} \times 5^{(2.4+2.6)} = 6^5 \times 5^5$
 $\Leftrightarrow 6^{(3.6+2x)} \times 5^5 = 6^5 \times 5^5 \Leftrightarrow 3.6 + 2x = 5 \Leftrightarrow 2x = 1.4$
 $\Leftrightarrow x = 0.7.$
58. Let $2^{3.6} \times 4^{3.6} \times 4^{3.6} \times 32^{2.3} = 32^x$.
Then, $2^{3.6} \times (2^2)^{3.6} \times (2^2)^{3.6} \times (2^5)^{2.3} = (2^5)^x$
 $\Leftrightarrow 2^{3.6} \times 2^{(2 \times 3.6)} \times 2^{(2 \times 3.6)} \times (2^5)^{2.3} = (2^5)^x$
 $\Leftrightarrow 2^{(3.6+7.2+7.2)} \times (2^5)^{2.3} = (2^5)^x \Leftrightarrow 2^{18} \times (2^5)^{2.3} = (2^5)^x$
 $\Leftrightarrow (2^5)^{3.6} \times (2^5)^{2.3} = (2^5)^x \Leftrightarrow (2^5)^{(3.6+2.3)} = (2^5)^x$
 $\Leftrightarrow (2^5)^{5.9} = (2^5)^x \Leftrightarrow x = 5.9.$
59. Let $3^{3.5} \times (21)^2 \times (42)^{2.5} \div 2^{2.5} \times 7^{3.5} = 21^x$.
Then, $3^{3.5} \times 7^{3.5} \times (21)^2 \times (21 \times 2)^{2.5} \div 2^{2.5} = 21^x$
 $\Leftrightarrow (21)^x = (3 \times 7)^{3.5} \times (21)^2 \times (21)^{2.5} \times 2^{2.5} \div 2^{2.5}$